Module 2 – Mernstack – HTML

• HTML Basics

Theory Assignment:

Question 1: Define HTML. What is the purpose of HTML in web development?

HTML (HyperText Markup Language) is the standard language used to create and structure content on the web. It defines the elements and structure of a web page using a system of tags, attributes, and content.

Purpose:

The purpose of HTML in web development is to structure and organize web content using a system of tags and elements. It defines the layout of a webpage, enabling the integration of text, images, links, and multimedia.

Question 2: Explain the basic structure of an HTML document. Identify the mandatory tagsand their purposes?

The basic structure of an HTML document includes the following mandatory tags:

1. <!DOCTYPE html>: Declares the document type and version of HTML being used.

2. <html>: The root element that encompasses the entire HTML document.

3. <head>: Contains metadata, such as the document title, character encoding, and links to stylesheets or scripts.

4. <title>: Specifies the title of the document, displayed on the browser tab.

5. <body>: Contains the visible content of the webpage, including text, images, links, and other elements.

Question 3: What is the difference between block-level elements and inline elements inHTML? Provide examples of each.

- **Block-Level Elements**: Start on a new line and take up the full width of their container. Used for structuring larger sections of a webpage.
  Examples: , <p>, <h1>, <ul>, <table>.
- **Inline Elements**: Do not start on a new line and only take up as much width as their content requires. Used within block-level elements for styling or grouping.
  Examples: , <a>, <img>, , .

Question 4: Discuss the role of semantic HTML. Why is it important for accessibility andSEO? Provide examples of semantic elements?

Role of Semantic HTML:

Semantic HTML refers to the use of HTML tags that convey the meaning and structure of content. These tags clearly define the role and purpose of the elements, making the content more understandable for browsers, search engines, and assistive technologies.

Importance for Accessibility and SEO:

1. Accessibility:

- Semantic tags make web content easier to navigate for users with disabilities who rely on screen readers or other assistive technologies.

2. SEO (Search Engine Optimization):

- Search engines use semantic tags to understand the structure and importance of content on a webpage.
- Proper use of semantic HTML improves search engine rankings and visibility.

Examples of Semantic Elements:

1. <header>: Represents the introductory content or a section heading.

2. <nav>: Defines a navigation menu.

3. <main>: Highlights the primary content of a webpage.

4. <section>: Groups related content together.

5. <article>: Represents independent, self-contained content like blog posts or news articles.

6. <footer>: Contains footer information for the page or section.

8. <aside>: Represents content related to the main content, such as sidebars.

Lab Assignment:

• Task: Create a simple HTML webpage that includes:

- A header (), footer (), main section (), and aside section ().
- A paragraph with some basic text.
- A list (both ordered and unordered).
- A link that opens in a new tab

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Simple HTML Webpage</title>
</head>
<body>
    <header>
        <h1>Welcome to My Webpage</h1>
        <p>Your one-stop destination for all things fun and informative!</p>
    </header>

    <aside>
        <h2>Quick Links</h2>
        <ul>
            <li><a href="https://example.com" target="_blank">Visit Example Site</a></li>
            <li><a href="https://anotherexample.com" target="_blank">Another Example</a></li>
        </ul>
    </aside>

    <main>
        <h2>About This Page</h2>
        <p>This is a simple HTML webpage designed to showcase basic HTML elements and structure.</p>
        <h3>Features:</h3>
        <ol>
            <li>Header and Footer</li>
            <li>Main and Aside Sections</li>
            <li>Lists and Links</li>
        </ol>
        <p>Check out the links in the aside section for more resources.</p>
    </main>

    <footer>
        <p>&copy; 2024 My Simple Webpage. All rights reserved.</p>
    </footer>
</body>
</html>
```

Output:

Welcome to My Webpage

Your one-stop destination for all things fun and informative!

Quick Links

- Visit Example Site
- Another Example

About This Page

This is a simple HTML webpage designed to showcase basic HTML elements and structure.

Features:

1. Header and Footer
2. Main and Aside Sections
3. Lists and Links

Check out the links in the aside section for more resources.

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• HTML Forms

Theory Assignment:

Question 1: What are HTML forms used for? Describe the purpose of the input, textarea, select, and buttonelements?

HTML Forms:

HTML forms are used to collect and submit user input to a server. They enable interaction between the user and the web application, allowing data to be captured and processed for tasks like login, registration, feedback, or search.

Purpose of Elements:

1. <input>:

o Used to collect user data like text, email, password, etc.

o Example:<input type="text" placeholder="enter your name ">.

2. <textarea>:

o Used for multi-line text input, such as comments or feedback, it also provide attributes like rows and cols

o Example: <textarea placeholder="enter your message" rows="5" cols="3"></textarea>

3. <select>:

- **o** Creates a dropdown menu to choose from multiple options.
- **o** Example:

```
<select>
    <option value="1">option 1</option>
    <option value="2">option 2</option>
</select>
```

4. <button>:

- **o** Triggers form submission or other actions when clicked.
- **o** Example:

```
<button type="submit">submit</button>
<button type="reset">reset</button>
```

Question 2: Explain the difference between the GET and POST methods in form submission. When should each be used?

GET	POST
1) In case of Get request, only **limited amount of data** can be sent because data is sent in header.	In case of post request, **large amount of data** can be sent because data is sent in body.
2) Get request is **not secured** because data is exposed in URL bar.	Post request is **secured** because data is not exposed in URL bar.
3) Get request **can be bookmarked.**	Post request **cannot be bookmarked.**
4) Get request is **idempotent** . It means second request will be ignored until response of first request is delivered	Post request is **non-idempotent.**
5) Get request is **more efficient** and used more than Post.	Post request is **less efficient** and used less than get.

When be used

GET Method:

- Used for retrieving data without modifying the server (e.g., search, filtering).
- Suitable for non-sensitive data and bookmarking URLs**.**

POST Method:

- Used for submitting data that changes server data (e.g., login, form submission).
- Ideal for sensitive information and file uploads.

Question 3: What is the purpose of the label element in a form, and how does it improve accessibility?

Purpose of the element:

The <label> element is used to define a label for form controls (e.g.<input>, <textarea> , <select> ,). It links descriptive text to a specific form element, making the form easier to understand and interact with.

How it improves accessibility:

1. Screen Reader Support: Screen readers can identify the purpose of the form control based on the associated label, improving usability for visually impaired users.
2. Clickable Area: When a is properly linked to a form element, clicking the label also activates the form control, enhancing usability.
3. Clear Context: Provides a clear and descriptive context for form fields, reducing confusion for all users.

Lab Assignment:

Task: Create a contact form with the following fields:

• Full name (text input) • Email (email input)

• Phone number (tel input)

• Subject (dropdown menu)

• Message (textarea)

• Submit button

Additional Requirements:

• Use appropriate form validation using required, minlength, maxlength, and pattern.

• Link form labels with their corresponding inputs using the for attribute

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Contact Form</title>
</head>
<body>
    <h1>Contact Us</h1>
    <form action="/submit-form" method="POST">
        <!-- Full Name -->
        <div>
            <label for="full-name">Full Name:</label>
            <input type="text" id="full-name" name="full-name" required minlength="2" maxlength="50">
        </div>

        <!-- Email -->
        <div>
            <label for="email">Email:</label>
            <input type="email" id="email" name="email" required>
        </div>

        <!-- Phone Number -->
        <div>
            <label for="phone">Phone Number:</label>
            <input type="tel" id="phone" name="phone" pattern="\\d{10}" required>
            <small>Format: 10 digits (e.g., 1234567890)</small>
        </div>

        <!-- Subject -->
        <div>
            <label for="subject">Subject:</label>
            <select id="subject" name="subject" required>
                <option value="">--Please Choose--</option>
                <option value="general">General Inquiry</option>
                <option value="support">Support</option>
                <option value="feedback">Feedback</option>
            </select>
        </div>

        <!-- Message -->
        <div>
            <label for="message">Message:</label>
            <textarea id="message" name="message" required minlength="10" maxlength="500"></textarea>
        </div>

        <!-- Submit Button -->
        <div>
            <button type="submit">Submit</button>
        </div>
    </form>
</body>
</html>
```

Output:

Contact Us

Full Name:
Email:
Phone Number: Format: 10 digits (e.g., 1234567890)
Subject: --Please Choose--
Message:
Submit

• **HTML Tables**

Theory Assignment:

Question 1: Explain the structure of an HTML table and the purpose of each of the following elements: <table>, <tr>, <th>, <td>, and <thead>?

Structure of HTML Table:

An HTML table is a structured way to display tabular data in rows and columns. It uses a combination of specific elements to define the table's structure and content**.**

Purpose of Elements:

<table>:

- The <table> element is the container for all the table content.
- It defines the boundaries of the table and organizes its rows, columns, and content.

<tr> (Table Row):

- The <tr> element defines a single row within a table.
- It acts as a container for table cells (<td> or <th>).

<th> (Table Header):

- The <th> element defines a header cell in a table.
- It is typically used at the top of a column or the start of a row to label the data within that row or column.

<td> (Table Data):

- The <td> element represents a standard data cell within a table.
- It contains the actual data displayed in the rows and columns of the table.

<thead> (Table Head):

- The <thead> element groups one or more <tr> elements that define the header section of a table.
- It helps to semantically separate the table's header rows from its body and footer.

Example of Table:

```
<table border="1">
  <thead>
  <tr>
  <th>Product</th>
  <th>Price</th>
  <th>Quantity</th>
  </tr>
  </thead>
  <tbody>
  <tr>
  <td>Iphone</td>
  <td>50,000</td>
  <td>1</td>
  </tr>
  <tr>
  <td>Samsung</td>
  <td>75,000</td>
  <td>1</td>
  </tr>
  </tbody>
 </table>
```

Explanation of Example:

<table>: Defines the table.

<thead>: Groups the header row (<tr> with <th> elements).

<tr>: Creates rows.

<th>: Creates bold, centered header cells ("Product", "Price", "Quantity").

<td>: Defines the data cells ("Iphone", "50,000", "1","Samsung","75,000","1").

Question 2: What is the difference between colspan and rowspan in tables? Provide examples?

Difference Between colspan and rowspan:

1. colspan:

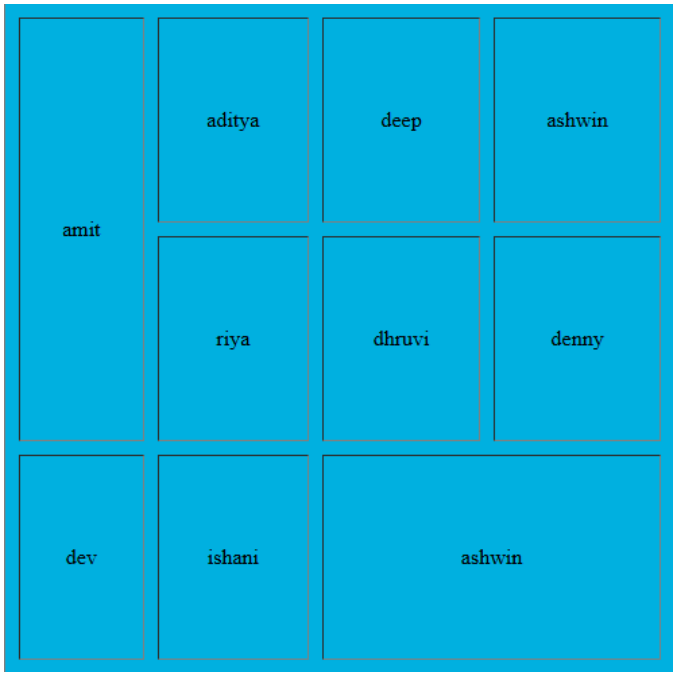
- Merges a cell across multiple columns.
- Expands a single cell horizontally to occupy the space of two or more columns.
- When a single value needs to cover multiple columns, such as in table headings or when grouping data.

2. rowspan:

- Merges a cell across multiple rows.
- Expands a single cell vertically to occupy the space of two or more rows.
- When a single value applies to multiple rows, such as a common label or grouping.

```html
<html>
    <head>
        <title>create a table </title>
        <body>
        <table border="2"align="center" bgcolor="sky blue" cellspacing="10" cellpadding="10" height="500px" width="500px">
            <tr align="center">
                <td rowspan="2">amit</td>
                <td>aditya</td>
                <td>deep</td>
                <td>ashwin</td>
            </tr>
            <tr align="center">
                <td>riya</td>
                <td>dhruvi</td>
                <td>denny</td>
            </tr>
            <tr align="center">
                <td>dev</td>
                <td>ishani</td>
                <td colspan="2">ashwin</td>
            </tr>
        </table>
        </body>
    </head>
</html>
```

Output:



Question 3: Why should tables be used sparingly for layout purposes? What is a better alternative?

Why Tables Should Be Used Sparingly for Layout Purposes:

1. Lack of Flexibility: Tables are rigid and not responsive, making it hard to adapt to different screen sizes and devices.

2. Accessibility Issues: Using tables for layout confuses screen readers, which are designed to interpret tables as data structures. This can hinder accessibility for users with disabilities.

3. Code Complexity: Tables for layout require excessive HTML, making the code harder to read, maintain, and debug.

4. Poor SEO: Search engines may misinterpret table layouts, negatively impacting rankings.

Better Alternative: CSS for Layout

CSS (Cascading Style Sheets) is a more efficient and modern way to create layouts. It provides flexibility, responsiveness, and accessibility.

Recommended CSS Techniques:

1. Flexbox:

- Ideal for creating one-dimensional layouts (e.g., rows or columns).
- Example:

```
<div style="display: flex;">
<div>Column 1</div>
<div>Column 2</div> </div>
```

2. CSS Grid:

- Best for two-dimensional layouts, such as grids or complex designs.
- Example:

```
<div style="display: grid; grid-template-columns:1fr 1fr;">
<div>Item 1</div>
<div>Item 2</div>
</div>
```

Conclusion:

Instead of tables, use CSS Flexbox or CSS Grid for creating layouts. They are more responsive, accessible, and maintainable while adhering to modern web development standards.

Lab Assignment:

- **Task: Create a product catalog table that includes the following columns:**
- **Product Name**
- **Product Image (use placeholder image URLs)**
- **Price**
- **Description**
- **Availability (in stock, out of stock)**

Additional Requirements:

- **Use thead for the table header.**
- **Add a border and some basic styling using inline CSS.**
- **Use colspan or rowspan to merge cells where applicable.**

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h1>Product Catlog</h1>
    <table border="1" cellspacing="0" cellpadding="10" style="border-collapse: collapse; width:100%; text-align: left;">

        <thead style="background-color: #f4f4f4;">
            <tr>
                <th>Product Name</th>
                <th>Product Image</th>
                <th>Price</th>
                <th>Description</th>
                <th>Avaliability</th>
            </tr>

        </thead>
    <tbody>
            <tr>
                <td rowspan="2">Product a</td>
                <td rowspan="2" align="center">
                    <img src="photo-1523275335684-37898b6baf30.avif" alt="Product a" style="width: 200px; height: auto;">
                </td>
                <td>30$</td>
                <td>It Is Best For Man</td>
                <td>In Stock</td>
            </tr>
            <tr>
                <td colspan="3" style="text-align: center; background-color: #f9f9f9;">
                    Buy 1 Get 1 Free
                </td>
            </tr>
            <tr>
                <td>Product b</td>
                <td align="center">
                    <img src="download.jpg" alt="product b" style="width: 200px; height: auto;">
                </td>
                <td>1500$</td>
                <td>It Is Good For Photography </td>
                <td>In Stock</td>
            </tr>
            <tr>
                <td>product c</td>
                <td align="center">
                    <img src="images.jpg" alt="product c" style="width: 200px; height: auto;">
                </td>
                <td>50$</td>
                <td>It is classic</td>
                <td>Out Of Stock Shortly</td>
            </tr>
            </tbody>
    </table>
    </body>
</html>
```

Output:

Product Catlog

Product Name	Product Image	Price	Description	Availability
Product a		30$	It Is Best For Man	In Stock
		Buy 1 Get 1 Free		
Product b		1500$	It Is Good For Photography	In Stock
product c		50$	It is classic	Out Of Stock Shortly